

CSCE 632 ACCESSIBLE COMPUTING · MODULE 07 · LECTURE NOTES

The Standard

WCAG's structure, and what a conformance claim means.

MODULE OUTCOMES

Explain WCAG's structure and the POUR principles.

Explain the conformance levels and their basis.

State what a conformance claim establishes.

Explain what a standard can and cannot encode.

Use the standard as a floor rather than a target.

1 The structure

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4 PRINCIPLES -- POUR
  Perceivable    available to a sense you have
  Operable       usable with the inputs you have
  Understandable content and behaviour both
  Robust         works with assistive technology,
                  now and in the future

THEN GUIDELINES under each principle
  goals, not testable -- they organise the criteria

THEN SUCCESS CRITERIA under each guideline
  testable statements, pass or fail
  each at level A, AA, or AAA
  technology-independent by design

THEN TECHNIQUES, which are NOT normative
  sufficient techniques: one way to pass
  failures: known ways to fail
  and you may pass by other means entirely

SO: criteria are the requirement, techniques are
advice. Confusing the two is the commonest error in
using the document.
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Criteria are the requirement and techniques are advice — confusing the two is the commonest error in using the document, and it goes in both directions: people treat a sufficient technique as mandatory (it is one way, not the way) and people treat the absence of a technique as permission (the criterion still applies). **The criteria-versus-techniques distinction is the practical key to the standard**, and knowing it makes the document far less intimidating.

AND POUR IS A GOOD DECOMPOSITION, WHICH IS WORTH SAYING BECAUSE MOST ACRONYMS ARE NOT

Perceivable, Operable, Understandable, Robust map onto genuinely different failure modes — can the information reach a sense you have; can you drive the interface with the inputs you have; can you follow the content and the behaviour; does it survive your particular assistive technology.

And they are roughly ordered by consequence — an imperceptible interface is a total failure and an understandability problem is usually a partial one — which makes the order a useful triage sequence when you have more defects than time.

Plus Robust is the one people skip and is the one that ages — it is the semantics requirement: markup that correctly declares what things are keeps working with assistive technologies nobody has written yet. **It is why Module 08 exists,** and it is the principle with the longest shelf life.

So the principles are worth using as a diagnostic frame: given a defect, ask which letter it breaks, which usually identifies the fix's category immediately — a Perceivable failure wants an alternative, an Operable one wants an input path, a Robust one wants semantics.

2 The levels

Level	What it actually means	In practice
A	The minimum. Its absence blocks entire groups of users completely.	Not optional, and not sufficient either — an A-conformant page can still be very hard to use.
AA	Achievable across content types without a fundamental redesign or a change of medium.	The usual legal and procurement target, and the level almost every policy names.
AAA	Not achievable for all content — some criteria are impossible for some media (sign interpretation of live audio, for instance).	Target it where it applies to your content; it is not claimable site-wide and the standard says so.

The levels are about achievability, not about importance — a AAA criterion is not a nicety, it is one that cannot reasonably be required of every kind of content. The enhanced contrast ratio at AAA matters a great deal to the people who need it; it sits at AAA because requiring 7:1 everywhere would constrain design more than the standard was willing to. **The achievability-not-importance point is widely misunderstood and changes how you read the levels:** AAA is a list of things to do where you can, not a list of things that barely matter.

AND CONFORMANCE IS ALL-OR-NOTHING PER PAGE, WHICH HAS CONSEQUENCES

A page conforms at a level only if every criterion at that level and every level below it is met — there is no partial credit, and one failure fails the page, which is a stricter rule than most people assume.

And a complete process must conform end to end — a conformant checkout with one inaccessible step is a non-conformant checkout, because the conformance unit is the whole process — **which is the full-pages-and-complete-processes requirement** and is the part most often ignored.

Which makes honest claims hard and dishonest ones easy — 'AA conformant' is asserted very much more often than it is true, and Module 09 §1 explains how that happens without anybody intending to lie: an automated tool returns green, and green is read as conformant.

So a claim should name what was tested, how, and when — which is Module 13 §2's form and is what a serious accessibility statement actually contains, including the known exceptions.

3 What it establishes

- **It says: these specific testable statements were checked and passed — which is real information and is genuinely worth having**, and is more than most software can say about any of its quality attributes.
- **It does not say the interface is usable — every single criterion can pass while the task remains impractical, which Module 09 §3 demonstrates with concrete examples** (a conformant but unusable form, a correctly labelled but hopeless navigation).
- **It does not cover cognitive accessibility adequately**, for **Module 06 §1's reasons** — so conformance is least informative for the largest population.
- **And it says nothing whatever about whether anybody with a disability was involved in the testing — which is the question Module 09 §4 is about and the one that matters most**, and which no conformance level requires.
- **So treat it as a floor: necessary, checkable, insufficient — which is the honest three-word summary of what a conformance claim is worth**, and is not a dismissal: a floor is load-bearing.

4 What a standard can encode

A STANDARD CAN ONLY REQUIRE WHAT CAN BE TESTED THE SAME WAY BY TWO PEOPLE

Which is why contrast has a number and simplicity does not — the inter-rater testability requirement is what makes a standard enforceable, and it is also what determines its coverage: anything two careful people would score differently cannot be a criterion.

So the gaps are not oversights — they are precisely the requirements that resisted operationalisation, which is Module 06 §1's gap seen from the standard's side rather than from the population's.

And this generalises well beyond accessibility: any metric-driven process is shaped by what is measurable rather than by what matters — which is CSCE 671 Module 12 §4's argument and CSCE 676 Module 13's proxy problem, arriving in a third setting and with the same structure each time.

So use the standard for what it is good at — the mechanical, checkable, arguable requirements, which it encodes extremely well — and do not mistake the covered set for the required set, which is the error the rest of this course is arranged to prevent.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
WCAG 2.2, with the quick reference (free)	The document itself — and read the structure before any individual criterion, because the structure is the navigation.
Understanding WCAG 2.2 (free)	§§1 and 2 — the intent behind each criterion, which is where the reasoning lives and the part worth actually reading.
WCAG 2.2 conformance requirements (free)	§2's callout — the five conformance requirements, including full pages and complete processes.
The W3C's accessibility statement generator (free)	§2's honest claim — what a statement should contain, including known exceptions and a contact route.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **State POUR** and give one failure under each principle.
- 2 **Classify ten defects** from Project 1 by principle.
- 3 **Find a sufficient technique** and a different way of passing the same criterion.
- 4 **Explain the levels to somebody** without saying that AAA is less important.
- 5 **Find a AAA criterion** that matters greatly to the people who need it.
- 6 **Take one multi-step process** and test every step, not only the main page.
- 7 **Find a published 'AA conformant' claim** and test three criteria on it.
- 8 **Write an honest accessibility statement** for something you built.
- 9 **Name three requirements from Module 06** that no criterion encodes, and say why.
- 10 **State the general lesson about standards** in your own words.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Give WCAG's four structural levels.
- 2 What is the difference between a criterion and a technique?
- 3 State POUR, with a failure mode for each.
- 4 Why is Robust the one that ages well?
- 5 What do the conformance levels measure?
- 6 Why is AAA not claimable site-wide?

- 7 State the two all-or-nothing rules.
- 8 What does a conformance claim establish, in three words?
- 9 Name three things it does not establish.
- 10 What can a standard require, and what follows about its gaps?